

Chapter 37 - The Terminal Serial Stream

The Intuition Engine has one character-serial port, called the terminal. It is the path BASIC uses for PRINT, GET, and INPUT; it is also the path that a freshly-booted machine uses to present its startup banner and the prompt. From a program's point of view it is a byte stream in (keyboard) and a byte stream out (console).

Every register lives in the terminal / serial block (0xF0700-0xF07FF).

37.1 Output: a single write-only port

Address	Name	R/W	Meaning
0xF0700	output byte	W	Writing a byte sends one character to the terminal. Reading returns 0.

There is no buffer status to wait on for output. Writing 0xF0700 enqueues the byte immediately. ASCII 0x0A ends a line; ASCII 0x0D is interpreted as carriage return. Bytes outside the printable ASCII range are sent verbatim and may be rendered as glyphs in the current font (see Appendix B) or dropped, depending on the terminal mode.

Two aliases exist for short-form addressing:

Address	Origin	Meaning
0xF700	16-bit absolute on M68K (.W addressing) - sign-extended below	Always reaches 0xF0700 on M68K.
0xFFFFF700	32-bit sign-extended form of 0xF700	Same byte, same effect.

The 6502 and Z80 do **not** see 0xF700 as the terminal output register: their CPU adapters intercept that address as a bank-control register before the bus translation runs. Bare 8-bit code that needs to print uses the cooked terminal stream via its bank-windowed view of 0xF0700, the ULA (Chapter 8), the VGA (Chapter 5), or one of the other display engines.

37.2 Output status

Address	Name	R/W	Meaning
0xF0704	terminal status	R	Bit 0 = at least one input byte is available; bit 1 = output ready.

Bit 1 of the status register is always set: the terminal output side never stalls. Bit 0 tells a program whether reading the input register will return a real byte or zero.

37.3 Input: cooked stream

Address	Name	R/W	Meaning
0xF0708	input dequeue	R	Read one input byte, then advance the input queue. Returns 0 if the queue is empty.
0xF070C	line status	R	Bit 0 is set when at least one complete line (ending in 0x0A) is queued.

The cooked input stream is what INPUT reads. Characters are delivered to it as the user types them, subject to the line-input flag in section 37.5.

A program that wants raw single-key reads (no line buffering, no waiting for Enter) uses the cooked-key path at 0xF0728 / 0xF072C described in Chapter 36 instead.

37.4 Local echo

Address	Name	R/W	Meaning
0xF0710	echo flag	R/W	Bit 0 = local echo. Default 1.

When local echo is on, the terminal prints each input character at the cursor as the user types. Programs that read the input register and want to do their own echo (a password prompt, a single-character menu) write 0 to clear bit 0. Writing 1 turns echo back on.

37.5 Line-input mode

Address	Name	R/W	Meaning
0xF0724	terminal control	R/W	Bit 0 = line-input mode. Default 1.

In line-input mode (the default) keystrokes accumulate in the cooked input queue at 0xF0708 until the user types Enter, which appends 0x0A and unblocks any reader that was waiting for a complete line. In single-character mode (bit 0 cleared) each key is delivered to the cooked-key queue at 0xF0728 instead, without buffering.

BASIC's INPUT uses line mode; BASIC's GET uses single-character mode. Mixed-mode programs flip the flag explicitly before each read.

37.6 The sentinel

Address	Name	R/W	Meaning
0xF07F0	sentinel	W	Writing the exact value 0xDEAD stops the current CPU.

The sentinel is a way for a running program to stop itself cleanly. The exact value 0xDEAD is required; any other value is ignored. The stop takes effect when the write completes; the CPU that issued it does not see the next instruction.

In normal programs the sentinel is reserved for test drivers and self-terminating image executor payloads. BASIC has no keyword that writes it.

37.7 Reserved cursor and colour registers

Address	Name	Status
0xF0714	cursor X	Reserved for future use. Reads return 0; writes are ignored.
0xF0718	cursor Y	Same.
0xF071C	fg colour	Same.
0xF0720	bg colour	Same.

These addresses are part of the terminal block's address map but are not implemented in the current Intuition Engine. They are documented so that programs do not accidentally collide with them.

37.8 A character at a time

The smallest useful interaction:

```

li      r2, #'?'
store.b r2, 0xF0700      ; print '?'

.wait:
load.l  r1, 0xF0704
li      r3, #1
and     r1, r1, r3
beqz    r1, .wait
load.l  r1, 0xF0708      ; consume the input byte

```

This is the building block for every text-mode program on the Intuition Engine: print a prompt, poll the status bit, read one byte. Everything in BASIC's INPUT, every dialog box drawn by a machine-code program, and every transcript captured during debugging is layered on top of this one byte-stream port.

37.9 What this chapter is not

The terminal is the only serial port the Intuition Engine has. There is no UART, no parallel port, no IEC bus, no RS-232 line, and no separate keyboard scan port. Programs that need binary data move it through the File I/O block (Chapter 34) or through the cross-CPU coprocessor channel (Chapter 31). Programs that need to receive raw keystrokes use the input registers in Chapter 36.